

Mark Jennings

Applied Roboticist | markjennings97@gmail.com | <https://makr.org>

Work Experience	Education
Elevate Robotics <i>Robotics Engineer, September 2024 – Present</i> • Prototyped a series of lightweight, high-payload robotic manipulators with passive gravity balancing • Integrated payload detection and halting functionality with load cell and custom fail-safe braking system • Led mechanical analysis and technical writing for NASA SBIR contract, successfully secured next phase for \$850k	MS Mechanical Engineering <i>UT Austin, Aug. 2019 – Aug. 2021</i> • Research thesis: <i>Manipulator Control in Collaborative Assembly</i>
Los Alamos National Laboratory <i>R&D Engineer, October 2021 – August 2024</i> • Overhauled nuclear glovebox with the first autonomous robotic arm in US plutonium part production • Developed control software, operating procedures, and tooling for multiple nuclear manufacturing processes • Coordinated intern program	BS Mechanical Engineering <i>UT Austin, Aug. 2015 – May 2019</i>
Nuclear and Applied Robotics Group <i>Graduate Research Assistant, August 2019 – August 2021</i> • Developed a C++ package to augment assembly tasks with a collaborative robot via force-sensing and admittance control, reducing reported worker physical effort by 57%	Skills
Sandia National Laboratory <i>R&D Intern, Summer 2019</i> • Designed additively manufactured metal components and developed corresponding qualification standards • Led 1st place intern team in design competition	Mechanical: • CAD: Onshape, SolidWorks, etc. • Fabrication: Additive manufacturing, manual machining/turning, CNC, sheet metal bending, laser cutting, die casting, injection molding, vacuum forming, extrusions, etc. • Hardware: Robotic actuators, motor drivers, force/torque sensors, load cells, encoders, brakes, etc. • Analysis: FEA, tolerance stackup, actuator selection, bushing/bearing selection, FMEA
Apptronik <i>Mechanical Engineering Intern, Summer 2018</i> • Derived forward kinematic equations for an advanced humanoid bipedal robot • Updated actuator testbed product to achieve higher payloads with lower fabrication costs	Software: • Languages: C/C++, Python • Other: Linux, Git, ROS, various sims
Rehabilitation and Neuromuscular (ReNeu) Robotics Lab <i>Undergraduate Research Assistant, May 2016 – May 2019</i> • Designed and fabricated components and actuators for rehabilitation exoskeleton robots	Outreach
	Los Alamos FIRST Tech Challenge <i>Mentor/Coach, 2022 – 2024</i> • Taught middle schoolers STEM, problem-solving, and teamwork
	UT Robotics & Automation Society <i>Mentor/Officer, 2015 – 2019</i> • Mentored first-year robotics teams and held several leadership roles